

DotNetNuke arbitrary code execution vulnerability (CVE-2017-9822) analysis advisory

DotNetNuke任意コード実行(CVE-2017-9822)分析 - Author not stated, cert.360.cn.

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DotNetNuke arbitrary code execution vulnerability (CVE-2017-9822) analysis advisory - 360CERT

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0x00 Background

DNN uses web cookies to identify users. A malicious user can decode one of such cookies and identify who that user is, and possibly impersonate other users and even upload malicious code to the server

--DNN security-center

On 5 July 2017, the DNN security section published a critical vulnerability with the identifier CVE-2017-9822. Later the vulnerability reporters Alvaro Muñoz (@pwntester) and Oleksandr Mirosh disclosed some of the details at BlackHat USA 2017. 360CERT followed up and analysed the vulnerability and the attack and exploitation scenario of using

XmlSerializer for serialization/deserialization in .net, and confirmed it as a critical vulnerability.

0x01 Vulnerability overview

DNNPersonalization is a Cookie in DNN used to store the personal data of users who are not logged in. This Cookie can be modified by an attacker to achieve attacks such as arbitrary file upload to the server and remote code execution.

0x02 Attack surface impact

1. Impact scope

Vulnerability level: critical

It is claimed that more than 750,000 users worldwide use DNN to build their websites, so the scope of impact is large.

2. Affected versions

All versions from 5.0.0 to 9.1.0

3. Fixed versions

DNN Platform 9.1.1 and EVOQ 9.1.1

0x03 Vulnerability details

1. Vulnerable code

PersonalizationController.cs lines 66-72:

[\[image: enter image description here\]](#)

After the value of DNNPersonalization is obtained from the Cookie, it is passed on to the DeserializeHashTableXml method in Globals.

Globals.cs lines 3687-3690:

[\[image: enter image description here\]](#)

Then follow into the DeSerializeHashtable method in XmlUtils.

XmlUtils.cs lines 184-218:

[\[image: enter image description here\]](#)

This method uses the value of the type attribute in the item element to set the type, and at line 208 here it deserializes the content of that element, which is exactly the trigger point of the vulnerability. The process from the controllable input point to the final exploitable place in the vulnerable code is fairly straightforward. Next comes the

exploitation analysis for vulnerable points like this that use XmlSerializer for deserialization.

0x04 Exploitation analysis

1. Use of XmlSerializer

[\[image: enter image description here\]](#)

When serializing or deserializing a class, the type information of that class must be passed in. Let us look at the form of the generated serialized data:

[\[image: enter image description here\]](#)

It is simply an XML document, where the class name and the member variables are represented as elements.

2. Building the exploit chain

Let us modify the TestClass class above and encapsulate its member variable test.

[\[image: enter image description here\]](#)

If we now observe the output of the code during deserialization, it is obvious that the setter is called automatically, so the setter can serve as the first step of the exploit chain. Next we have to go and find some classes that can be used for the attack.

System.Windows.Data.ObjectDataProvider can call any method of any class referenced at runtime. An example:

[\[image: enter image description here\]](#)

This is equivalent to calling TestClass.FuncExample("JustATest!"). The member variables in ObjectDataProvider are all encapsulated, and every time a setter is called it checks whether the parameters are sufficient; if they are, it automatically goes and calls the method that was passed in. The process is shown using a picture borrowed from the BlackHat talk.

[\[image: enter image description here\]](#)

In this way, if what is serialized is an ObjectDataProvider class, then during deserialization the effect of calling an arbitrary method can be achieved. Then it is enough to find another class that has a method achieving the desired exploitation effect. For example, in DNN there is a class that can achieve an arbitrary file upload effect, the PullFile method in DotNetNuke.Common.Utilities.FileSystemUtils:

[\[image: enter image description here\]](#)

3. Payload generation

To generate the payload there is one more problem to solve, namely that ObjectDataProvider contains a System.Object member variable (objectInstance), and at execution time XmlSerializer does not know the concrete type of this variable, so it cannot

serialize it. But this problem can be solved by using ExpandedWrapper to extend the type of the property.

[image: enter image description here]

The generated content is as follows:

[image: enter image description here]

DNN gets the value of the type attribute of item, then calls Type.GetType to obtain the type of the serialized data and then deserializes it. In that case the name of the corresponding assembly has to be added. The value of type can be obtained with the code below:

[image: enter image description here]

[image: enter image description here]

Combining this with the DNN code, the final Payload is generated:

[image: enter image description here]

0x05 Exploitation verification

Put the DeSerializeHashtable function where the vulnerability triggers into a local setup to do an exploitation verification.

[image: enter image description here]

Then look at the server side, and you can see that the exploitation succeeded.

[image: enter image description here]

0x06 Fix recommendations

360CERT recommends upgrading to the latest version, DNN Platform 9.1.1 or EVOQ 9.1.1.

0x07 Timeline

2017-7-5 The vendor published a security advisory and provided a fix update

2017-8-2 360CERT completed the analysis of the vulnerability and produced a report

0x08 References

<https://www.blackhat.com/docs/us-17/thursday/us-17-Munoz-Friday-The-13th-Json-Attacks.pdf>

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